Chapter 89 — Memory Management & Leak Prevention

> **Part X — Performance, Observability & Quality** · Complexity: **Complex** (≥40 interview questions)
>
> **Prerequisites:** Ch 44 (lifecycle), Ch 3 (threading/coroutines), Ch 59 (image loading), Ch 46 (DI/scopes), Ch 85 (perf monitoring), Ch 83 (crash/OOM). Pairs with Ch 90 (jank), Ch 88 (cold start), Ch 93 (lists).
>
> **Reference surface in StreamCart:** keeping the app's **memory** healthy — understanding the **managed heap + GC**, avoiding **leaks** (holding `Context` / `View` /lifecycle objects past their life, Ch 44), preventing **OutOfMemoryError** (large bitmaps, Ch 59; unbounded caches), responding to **memory pressure** (`onTrimMemory`), and **detecting** leaks (LeakCanary) and footprint (Android Studio Memory Profiler, field, Ch 85) — because leaks and bloat cause **jank** (GC pauses, Ch 90), **OOM crashes** (Ch 83), and background kills.

89.1 Problem Definition and Real-World Context

Android apps run under a **managed heap** with **garbage collection** and a **per-app memory limit**; exceeding it causes **OutOfMemoryError** (a crash, Ch 83), and churning it causes **GC pauses** that drop frames (**jank**, Ch 90). The two dominant problems are **memory leaks** — objects that should be freed but are **held** by a longer-lived reference (the classic: an `Activity` / `View` / `Context` captured by a static field, a singleton, a running coroutine, a registered listener, or an inner class that outlives its host, Ch 44) — and **memory bloat** — holding **too much** at once (giant bitmaps, Ch 59; unbounded caches/lists). **Memory management & leak prevention** is the discipline of understanding the heap/GC, **scoping** references to lifecycles (Ch 44), **bounding** caches, responding to `onTrimMemory` pressure, and **detecting** leaks (LeakCanary) and footprint (Memory Profiler + field, Ch 85). Leaks and bloat make the app slow (GC), crash (OOM), and more likely to be **killed** in the background (Ch 44).

Concretely, teams hit:

- **Context/View/Activity leaks (the #1 Android leak).** A **static** field, a **singleton** (Ch 46), a **companion object**, a long-running **coroutine/thread** (Ch 3), a registered **listener/callback/BroadcastReceiver** (Ch 44), or a non-static **inner class** captures an `Activity` / `Fragment` / `View` / `Context` and **outlives** it — so the whole view hierarchy can't be collected. Rotations make it worse (a new `Activity` each time).
- **Coroutine/flow leaks (Ch 3).** A coroutine launched in the wrong scope (`GlobalScope` , or a scope not tied to lifecycle) keeps running and **captures** the `Activity` / `ViewModel` . Must use **lifecycle/viewModelScope** (Ch 44) and collect with `repeatOnLifecycle` .
- **Bitmap/image bloat → OOM (Ch 59).** Bitmaps are the **largest** objects; loading full-res images into memory, not downsampling, or unbounded image caches causes **OOM**. Image loaders (Coil/Glide, Ch 59) manage this — but misuse (caching originals, huge `ImageView` s) still bloats.
- **Unbounded caches/collections.** An in-memory cache or list with **no eviction** grows until OOM. Caches must be **bounded** (`LruCache` , size/count limits) and respond to memory pressure.
- **Ignoring memory pressure.** The OS signals pressure via `onTrimMemory` / `ComponentCallbacks2` (and the deprecated `onLowMemory`); an app that ignores it holds memory the OS needs → the app is **killed** sooner (Ch

44). Must **release** caches/non-critical memory on trim.

- **GC churn → jank (Ch 90).** Allocating heavily **per frame** (in `onDraw`/recompose, Ch 90) causes frequent GC → **pauses** → dropped frames. Avoid allocations on the hot path.
- **Leaks found late / not at all.** Without **LeakCanary** (debug) and **Memory Profiler** leaks ship to production and show up as **OOM crashes** (Ch 83) and background kills — hard to diagnose after the fact. Need **detection** in dev + **field OOM** monitoring (Ch 85).
- **The `WeakReference` / lifecycle-scoping toolkit.** The fixes: **scope to lifecycle** (Ch 44), **unregister** listeners, **cancel** coroutines, use `WeakReference` where a back-reference is unavoidable, **don't** hold `Context`/`View` in long-lived objects (use `applicationContext` where appropriate), and **bound** caches.

The architecture: understand the **managed heap + GC + per-app limit**; **scope every reference to a lifecycle** (Ch 44) — cancel coroutines (`viewModelScope`/`lifecycleScope`, Ch 3), unregister listeners/receivers, never hold `Activity`/`View`/`Context` in singletons/statics (use `applicationContext`), use `WeakReference` for unavoidable back-refs; **bound** caches (`LruCache`) and image memory (Ch 59); **respond to `onTrimMemory`** by releasing non-critical memory (Ch 44); **avoid per-frame allocations** (GC churn → jank, Ch 90); and **detect** leaks (LeakCanary) + footprint (Memory Profiler) + **field OOM** (Ch 85/83). This chapter is about **keeping memory healthy so the app doesn't jank, OOM, or get killed**.

Three practical use cases

1. **Fix an Activity leak (StreamCart).** A singleton analytics helper (Ch 46) held an `Activity` `Context`; rotating leaked a view hierarchy each time. Fix: inject `applicationContext`, not the `Activity`; LeakCanary confirms the leak is gone.
2. **Bound an image cache → stop OOM (Ch 59).** The product gallery cached full-res bitmaps unbounded → OOM on low-RAM devices. Fix: Coil's bounded memory cache + downsampling to view size; respond to `onTrimMemory` by clearing it.
3. **Scope a coroutine (Ch 3).** A `GlobalScope.launch` outlived its screen and captured the `ViewModel`. Fix: `viewModelScope`/`repeatOnLifecycle` so it cancels with the lifecycle (Ch 44).

Why this is architectural

Memory management composes lifecycle (Ch 44), threading/coroutine scopes (Ch 3), image loading (Ch 59), DI scopes (Ch 46), and monitoring (Memory Profiler + field OOM, Ch 85/83). The decisions — lifecycle-scoped references, bounded caches, memory-pressure response, allocation discipline, leak detection — determine whether the app stays fast (no GC jank, Ch 90), stable (no OOM, Ch 83), and resident (not killed, Ch 44).

89.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Scope every reference to a lifecycle; cancel coroutines, unregister listeners (Ch 44/3).
F2	Never hold Activity/View/Context in singletons/statics; use applicationContext / WeakReference (Ch 46).
F3	Bound caches (LruCache) and image memory; downsample bitmaps (Ch 59).
F4	Respond to onTrimMemory by releasing non-critical memory (Ch 44).
F5	Avoid per-frame allocations (GC churn → jank, Ch 90).
F6	Detect leaks (LeakCanary) + footprint (Memory Profiler) + field OOM (Ch 85/83).

Non-functional requirements

Attribute	Policy
Stability	No OOM (bounded memory, Ch 83).
Smoothness	No GC jank (allocation discipline, Ch 90).
Residency	Respond to trim → not killed early (Ch 44).
Detectability	LeakCanary (dev) + field OOM (Ch 85).
Correctness	Lifecycle-scoped; no leaks.

Scope

- **In scope:** managed heap + GC + per-app limit, leak taxonomy (Context/View/Activity, coroutine/flow, listener/receiver, static/singleton, inner class), lifecycle scoping (Ch 44/3), `WeakReference` / `applicationContext`, bounded caches (`LruCache`) + image memory (Ch 59), `onTrimMemory` / `ComponentCallbacks2`, per-frame allocation discipline (Ch 90), detection (LeakCanary, Memory Profiler, field OOM, Ch 85/83).
- **Out of scope:** lifecycle detail (Ch 44), coroutine internals (Ch 3), image-loader internals (Ch 59), jank/rendering (Ch 90), crash reporting mechanics (Ch 83), DI (Ch 46).

Assumptions & constraints

- ART managed heap with per-app limit (varies by device RAM); GC is automatic. LeakCanary in debug; Memory Profiler for footprint; field OOM via vitals/Crashlytics (Ch 85/83). References **lifecycle-scoped** (Ch 44); caches **bounded**; `onTrimMemory` **honored**. `minSdk 24`.

89.3 System Design Fundamentals

- **Managed heap, GC, and the per-app limit.** ART manages a **heap** and **garbage-collects** unreachable objects; the app has a **per-process memory limit** (device-dependent — tens to hundreds of MB). Exceed it → `OutOfMemoryError` (crash, Ch 83). **Reachability** is the key: an object is kept alive as long as a **GC root** (static field, live thread, stack) reaches it through references. A **leak** is an object still **reachable** that **should** be gone — the GC can't collect it because something holds it.
- **The Android leak taxonomy (what holds what):**
  - **Static/singleton holds Activity/View/Context (Ch 46):** a `static` / `companion` /singleton field referencing an `Activity`, `View`, or non-application `Context` → the whole view hierarchy leaks. **Fix:** hold `applicationContext` (app-lived) or nothing; inject the right scope (Ch 46).
  - **Listener/callback/receiver not unregistered (Ch 44):** registering a listener/ `BroadcastReceiver` /observer that holds the `Activity` and **not unregistering** in the paired lifecycle callback → leak. **Fix:** unregister in `onStop` / `onDestroy` / `onPause` (Ch 44), or use lifecycle-aware components.
  - **Coroutine/thread captures lifecycle (Ch 3):** a coroutine in `GlobalScope` or a raw `Thread` / `Handler` captures the `Activity` / `ViewModel` and outlives it. **Fix:** `viewModelScope` / `lifecycleScope`; `repeatOnLifecycle` for collection; cancel on destroy (Ch 44).
  - **Non-static inner class / anonymous class:** holds an implicit reference to the outer (`Activity`). **Fix:** static nested class + `WeakReference`, or scope properly.
  - **`Handler` / `Runnable` with delayed messages:** a posted `Runnable` holds the `Activity` until it runs. **Fix:** remove callbacks on destroy; `WeakReference`.
- **Bitmaps and OOM (Ch 59). Bitmaps are the biggest memory objects** (width × height × 4 bytes for ARGB_8888). Loading full-res into memory, not **downsampling** to the view size, or **caching originals** causes **OOM**. **Image loaders** (Coil/Glide, Ch 59) handle downsampling, bounded memory/disk caches, and lifecycle cancellation — use them; don't decode bitmaps by hand into unbounded caches.
- **Bounded caches (`LruCache`).** Any in-memory cache or collection **must be bounded** — by **size** (bytes, e.g., a fraction of `maxMemory`) or **count** — with **LRU eviction** (`LruCache`). Unbounded caches are a leak-by-design → OOM. Size caches relative to available memory.
- **Memory pressure: `onTrimMemory` / `ComponentCallbacks2` (Ch 44).** The OS signals pressure with `onTrimMemory(level)` — e.g., `TRIM_MEMORY_RUNNING_LOW`, `TRIM_MEMORY_UI_HIDDEN` (app backgrounded), `TRIM_MEMORY_COMPLETE` (about to be killed). A well-behaved app **releases non-critical memory** (clear in-memory caches, free UI resources when hidden) on trim — this both **avoids OOM** and makes the app **less likely to be killed** (Ch 44). Honor it.
- **Allocation discipline → avoid GC jank (Ch 90).** Allocating heavily on the **hot path** — per frame in `onDraw`, per recomposition (Ch 90), in tight loops — causes frequent **GC**, whose **pauses drop frames** (jank). **Avoid allocations** in draw/recompose; reuse objects; be mindful of autoboxing, lambdas capturing, and temporary collections on the hot path (Ch 90).
- **Detection: LeakCanary + Memory Profiler + field OOM (Ch 85/83). LeakCanary** (debug-only) automatically detects **retained** `Activity` / `Fragment` / `View` / `ViewModel` after they should be gone and gives the **reference chain** (what's holding it) — the single best leak tool. **Android Studio Memory Profiler** shows **footprint**, allocations, and lets you capture/ compare **heap dumps**. **Field OOM** (Ch 85/83) — OOM crashes and `TRIM_MEMORY` kills in production — tells you the real-device reality. Use LeakCanary in dev, profile footprint, watch field OOM.

- `WeakReference` / `applicationContext` **(the escape hatches).** When a back-reference is **unavoidable** (a callback that must reference its host), use a `WeakReference` so it doesn't **pin** the host. When a long-lived object needs a `Context`, use `applicationContext` (app-lived, safe) — **never** an `Activity` / `View` `Context`. These, plus lifecycle scoping, are the toolkit.

Core insight: **Android runs a managed heap with GC and a per-app limit, so a leak is any object still reachable from a GC root (a static field, a live thread/coroutine, a registered listener) that should have been freed — most often an Activity/View/Context captured by a singleton or static (Ch 46), a coroutine in the wrong scope (Ch 3), a non-static inner class, or an unregistered listener/receiver that outlives its lifecycle (Ch 44); prevent leaks by scoping every reference to a lifecycle (viewModelScope/lifecycleScope + repeatOnLifecycle, Ch 3/44), unregistering listeners and canceling coroutines, never holding Activity/View/Context in long- lived objects (use applicationContext, or a WeakReference for unavoidable back-refs), bounding every cache (LruCache by size/count) and image memory with downsampling (Ch 59), and responding to onTrimMemory by releasing non-critical memory so the app avoids OOM (Ch 83) and is killed less eagerly (Ch 44); avoid per-frame allocations that churn the GC into frame-dropping pauses (Ch 90); and detect problems with LeakCanary (dev, gives the reference chain), the Memory Profiler (footprint and heap dumps), and field OOM/trim monitoring (Ch 85/83) — because leaks and bloat manifest as jank, OOM crashes, and background kills.**

89.4 High-Level Architecture


89.4 High-Level Architecture

Components & ownership

- **Heap/GC** keeps anything reachable from a GC root; a **leak** = wrongly reachable.
- **Fix leaks:** lifecycle scoping (Ch 44/3), `applicationContext` / `WeakReference` (Ch 46).
- **Fix bloat:** bounded caches + image downsampling (Ch 59); `onTrimMemory` response (Ch 44).
- **Allocation discipline** (no GC jank, Ch 90); **detection** (LeakCanary + Profiler + field, Ch 85/83).

Direction: reachable-but-should-be-gone = leak → scope/weak-ref; too-much = bloat → bound + trim; detect with LeakCanary + field OOM.

89.5 Detailed Component Design


89.5 Detailed Component Design

Component	Responsibility
`LifecycleScopedRef`	coroutines/listeners scoped + canceled with lifecycle (Ch 3/44)
`SafeContextHolder`	`applicationContext` / `WeakReference` — never hold Activity/View (Ch 46)
`BoundedCache`	`LruCache` size/count eviction; clear on trim
`MemoryPressureHandler`	`onTrimMemory` → release non-critical (Ch 44)
`LeakDetection`	LeakCanary (dev), Memory Profiler, field OOM (Ch 85/83)

89.6 End-to-End Data Flow

Leak lifecycle (create → hold → detect → fix)



Memory pressure response



89.7 Complete Kotlin Implementation

Package: `com.streamcart.memory`.

89.7.1 Lifecycle-scoped coroutines + collection (no leak, Ch 3/44)

```
class FeedViewModel @Inject constructor(private val repo: FeedRepository) : ViewModel() {
    // viewModelScope cancels on ViewModel clear - no leak, no outliving coroutine (Ch 3/44)
    fun refresh() = viewModelScope.launch(Dispatchers.IO) { repo.refresh() }
}

@Composable
fun FeedScreen(vm: FeedViewModel) {
    // collectAsStateWithLifecycle stops collecting when stopped - no leak (Ch 44)
    val state by vm.state.collectAsStateWithLifecycle()
    FeedList(state)
}
// NEVER: GlobalScope.launch { ... uses Activity/ViewModel ... }  // outlives lifecycle -> leak
```

89.7.2 Safe Context: applicationContext, never Activity (Ch 46)

```
@Singleton
class AnalyticsHelper @Inject constructor(
    @ApplicationContext private val appContext: Context,   // app-lived - SAFE (Ch 46); NOT an
Activity
) {
    fun log(event: String) { /* uses appContext - never leaks an Activity */ }
}
// BUG (leak): class Helper(val activity: Activity) held by a singleton -> whole view tree
leaks.
// If a back-ref to a short-lived host is unavoidable, hold a WeakReference (see 89.7.4).
```

89.7.3 Unregister listeners / receivers on lifecycle (Ch 44)

```
class LocationScreen : Fragment() {
    private val callback = LocationCallback { /* ... */ }
    override fun onStart() { super.onStart(); locationManager.register(callback) }   // register
    override fun onStop() { super.onStop(); locationManager.unregister(callback) }   //
UNREGISTER - no leak (Ch 44)
    // A registered callback holds this Fragment; not unregistering leaks it. Pair
register/unregister.
}
```

89.7.4 WeakReference for unavoidable back-ref + Handler cleanup

```
// Static nested class + WeakReference so a delayed Runnable doesn't pin the Activity:
class MainActivity : ComponentActivity() {
    private val handler = Handler(Looper.getMainLooper())
    private val task = WeakRunnable(this)
    override fun onDestroy() { handler.removeCallbacks(task); super.onDestroy() }   // remove -
no leak
    private class WeakRunnable(activity: MainActivity) : Runnable {
        private val ref = WeakReference(activity)                                    // weak -
doesn't pin
        override fun run() { ref.get()?.let { /* use if still alive */ } }
    }
}
```

89.7.5 Bounded cache + trim response (Ch 44/59)

```
@Singleton
class ThumbnailCache @Inject constructor() : ComponentCallbacks2 {
    private val maxKb = (Runtime.getRuntime().maxMemory() / 1024 / 8).toInt()       // ~1/8 of
heap
    private val cache = object : LruCache<String, Bitmap>(maxKb) {                   // BOUNDED
(Ch 59)
        override fun sizeOf(key: String, v: Bitmap) = v.byteCount / 1024
    }
    fun get(k: String) = cache.get(k); fun put(k: String, b: Bitmap) = cache.put(k, b)

    override fun onTrimMemory(level: Int) {                                          // memory
pressure (Ch 44)
        when {
            level >= ComponentCallbacks2.TRIM_MEMORY_RUNNING_LOW -> cache.evictAll() // release
non-critical
            level >= ComponentCallbacks2.TRIM_MEMORY_UI_HIDDEN -> cache.trimToSize(maxKb / 2)
        }
    }
    override fun onConfigurationChanged(c: Configuration) {}
    override fun onLowMemory() { cache.evictAll() }
}
// Prefer Coil/Glide's managed bounded memory cache (Ch 59) over hand-rolled bitmap caches.
```

89.7.6 Avoid per-frame allocations (GC jank, Ch 90)

```
// BAD: allocates every draw -> GC churn -> jank (Ch 90)
override fun onDraw(canvas: Canvas) {
    val paint = Paint()                     // allocation PER FRAME - avoid
    canvas.drawRect(RectF(0f,0f,w,h), paint)
}
// GOOD: hoist allocations out of the hot path
private val paint = Paint()                 // allocate ONCE
private val rect = RectF()
override fun onDraw(canvas: Canvas) { rect.set(0f,0f,w,h); canvas.drawRect(rect, paint) }  // no
per-frame alloc
// In Compose: hoist/remember objects; avoid allocating in the composition hot path (Ch 90).
```

89.7.7 Module structure

```
:core:memory  BoundedCache (LruCache + trim), SafeContextHolder, LeakDetection hooks
# Lifecycle scoping -> Ch 44/3; images -> Coil bounded cache (Ch 59); trim ->
ComponentCallbacks2 (Ch 44);
# detection -> LeakCanary (debug dep), Memory Profiler, field OOM (Ch 85/83). Allocation
discipline -> Ch 90.
```

89.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Coroutine outlives lifecycle	leak + crash on dead view	viewModelScope/lifecycleScope (Ch 3/44)
Listener not unregistered	Activity leak	pair register/unregister (Ch 44)
Static holds Context	view-tree leak	applicationContext / WeakReference (Ch 46)
Unbounded cache grows	OOM	LruCache size/count (Ch 59)
Per-frame allocation	GC jank	hoist allocations (Ch 90)
Ignoring trim	early kill / OOM	onTrimMemory release (Ch 44)
Bitmap not downsampled	OOM	decode to view size (Ch 59)
Rotation leaks accumulate	multiplying leaks	lifecycle-scoped; LeakCanary (Ch 44)

- **Lifecycle scoping is the master fix (Ch 44/3):** tying coroutines, listeners, and callbacks to `viewModelScope` / `lifecycleScope` / `repeatOnLifecycle` makes them **cancel/unregister automatically** when the owner dies — eliminating the most common leaks without manual cleanup.
- **Rotations compound leaks:** each config change creates a **new** `Activity`; a leak per rotation **multiplies** memory until OOM. LeakCanary catches retained activities after rotation.
- **Bounded + trim-responsive caches** keep footprint in check and make the app a **good OS citizen** — released memory on trim means **surviving longer** in the background (Ch 44).

89.9 Failure Scenarios and Recovery

Failure	Detection	Response	Outcome
Activity leak	LeakCanary chain	applicationContext / unregister / WeakReference (Ch 46/44)	freed
OOM crash	field OOM (Ch 83/85)	bound caches + downsample bitmaps (Ch 59)	stable
GC jank	Memory Profiler / frames (Ch 90)	hoist per-frame allocations	smooth
Killed early in bg	process-death metrics (Ch 44)	honor onTrimMemory	resident
Coroutine leak	LeakCanary / logs	lifecycle scope (Ch 3/44)	scoped
Cache grows unbounded	profiler footprint	LruCache eviction	bounded
Rotation leak pile-up	retained activities	lifecycle-scoped refs (Ch 44)	no accumulation

 89.9 Failure Scenarios and Recovery

> **Scope to lifecycle, bound everything, honor trim — and detect with LeakCanary.** The dominant Android memory failures are an Activity/View/Context leaked by a long-lived holder (lifecycle scoping + applicationContext/WeakReference fix it), OOM from bitmaps and unbounded caches (downsampling + LruCache fix it), and GC jank from per-frame allocations (hoisting fixes it). Respond to onTrimMemory to stay resident (Ch 44), and catch leaks in dev with LeakCanary + field OOM (Ch 85/83) before they crash production.

89.10 Scaling the Implementation

- **Level 1.** LeakCanary in debug; use `viewModelScope` / `collectAsStateWithLifecycle` (Ch 44); `applicationContext` in singletons (Ch 46).
- **Level 2.** Bounded caches (`LruCache`) + Coil-managed image memory (Ch 59); `onTrimMemory` response (Ch 44); unregister listeners.
- **Level 3 — production.** Field OOM monitoring (Ch 85/83); Memory Profiler in CI/manual audits; allocation discipline on hot paths (Ch 90); WeakReference where needed.
- **Level 4 — large.** Heap-dump analysis in CI (regression gates), memory budgets by device class, per-feature footprint tracking, large-object (bitmap) audits (Ch 59).
- **Level 5 — enterprise.** Memory SLOs, automated leak detection in CI (instrumented tests + LeakCanary), native-memory tracking, device-tiered memory policies, OOM-rate error budgets.

Avoid overengineering: lifecycle scoping + LeakCanary + bounded caches + `onTrimMemory` + Coil for images covers the vast majority. Don't hand-roll bitmap caches; don't hold `Context` in singletons; don't ignore trim; don't micro-optimize allocations without profiling.

89.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Reference scope	Lifecycle-scoped (Ch 44/3)	manual/unscoped	Lifecycle-scoped (auto cleanup)
Context in singleton	applicationContext (Ch 46)	Activity Context	applicationContext (no leak)
Back-ref	WeakReference	strong	Weak when unavoidable
Cache	Bounded LruCache (Ch 59)	unbounded	Bounded always
Images	Coil/Glide managed (Ch 59)	hand-decoded bitmaps	Managed loader
Pressure	Honor onTrimMemory (Ch 44)	ignore	Honor (residency + OOM)
Detection	LeakCanary + field (Ch 85)	none/late	LeakCanary + field

Lifecycle scoping vs manual cleanup: manual register/unregister and cancel is error- prone — one missed path leaks. **Lifecycle scoping** (`viewModelScope` , `repeatOnLifecycle` , lifecycle-aware components, Ch 44/3) makes cleanup **automatic** at the right time. Prefer scoping; use manual unregister only where no lifecycle-aware option exists.

applicationContext vs Activity Context in long-lived objects: holding an **Activity** `Context` in a singleton/static leaks the **entire view hierarchy** on every rotation. Holding **`applicationContext`** (app-lived) is safe for most needs (resources, system services). Use `applicationContext` in anything longer-lived than the `Activity` ; use the `Activity` context only for UI tied to that activity (themed inflation, dialogs) and don't retain it (Ch 46).

Managed image loader vs hand-rolled bitmap cache (Ch 59): hand-decoding bitmaps into a cache invites **OOM** (no downsampling, unbounded) and leaks (lifecycle). **Coil/Glide** downsample to view size, bound memory/disk caches, and cancel on lifecycle — far safer. Use a managed loader; don't hand-roll bitmap memory.

89.12 Security & Privacy Considerations

- **Clear sensitive data from memory (Ch 20/76):** tokens, keys, card data (Ch 76) held in memory should be **minimized and cleared** when no longer needed (and not swapped into caches/ logs). Memory bloat that retains sensitive data widens exposure (Ch 28).
- **Heap dumps may contain PII (Ch 28):** a captured heap dump can contain user data — handle **internally**, don't ship/share raw dumps; field OOM reports must be **PII-safe** (Ch 83/28).
- **Don't log memory contents:** OOM diagnostics are **footprint/metrics**, not memory contents (Ch 28).
- **Bounded caches reduce retention:** bounding caches also **limits how long** data lives in memory (privacy + stability).
- **No secrets in `WeakReference`-reachable singletons:** the escape-hatch holders are for lifecycle safety, not for stashing secrets (Ch 24).
- Client-only concern mostly: memory is a **client** property; the cross-cutting privacy note is keeping sensitive data minimal and out of heap dumps/logs (Ch 28).

89.13 Performance and Resource Optimization

- **Avoid allocations on the hot path (Ch 90)** — per-frame/per-recomposition allocations churn the GC → pauses → jank. Hoist/reuse objects (`Paint`, `RectF`, buffers).
- **Bound caches relative to `maxMemory`** — size by available heap, not a fixed number; evict (LruCache).
- **Downsample bitmaps to view size (Ch 59)** — the single biggest footprint lever; use a managed loader.
- **Release on trim (Ch 44)** — free memory when hidden/low → avoid OOM + stay resident.
- **Minimize retained scope** — don't keep large objects alive longer than needed; scope to lifecycle (Ch 44).
- **Profiling:** heap footprint (Memory Profiler), allocation rate on hot paths, GC frequency/ pauses (→ jank, Ch 90), retained set (LeakCanary), field OOM rate + trim kills (Ch 85/83).

89.14 Testing Strategy

Leaks are caught by LeakCanary (instrumented) and heap-dump assertions; bounds/trim are unit- testable.

```
class BoundedCacheTest {
    @Test fun `cache evicts beyond max size`() {
        repeat(1000) { cache.put("k$it", bitmap()) }
        assertTrue(cache.size() <= cache.maxSize())   // bounded (Ch 59)
    }
    @Test fun `onTrimMemory releases memory`() {
        cache.put("k", bitmap()); cache.onTrimMemory(TRIM_MEMORY_COMPLETE)
        assertEquals(0, cache.size())                 // released (Ch 44)
    }
}

@RunWith(AndroidJUnit4::class)
class LeakTest {   // LeakCanary / instrumented
    @Test fun `screen does not retain activity after finish`() {
        val scenario = launchActivity<MainActivity>()
        scenario.recreate(); scenario.recreate()       // rotations
        scenario.close()
        // LeakCanary fails the build on a retained Activity; or assert via heap dump (Ch 44)
    }
}
// StrictMode (debug) flags leaked Closeables/activities; Memory Profiler for footprint audits.
```

What to test: cache bounds + eviction (Ch 59); `onTrimMemory` release (Ch 44); no retained Activity after rotation/finish (LeakCanary); lifecycle-scoped coroutines cancel (Ch 3/44); StrictMode leak/closeable detection; footprint audits (Memory Profiler). Field OOM via vitals (Ch 85/83).

89.15 Production Observability

- **Metrics: OOM crash rate** (Ch 83), **memory footprint** percentiles by device class (field, Ch 85), `TRIM_MEMORY`**/low-memory kill** rate (Ch 44), GC pause frequency (→ jank, Ch 90), retained-instance reports (LeakCanary in dev; some field leak tracking).
- **Logs:** OOM/trim events — **PII-safe**, footprint not contents (Ch 28/83).
- **Debugging:** OOM → heap dump / top retained objects (bitmaps? caches?); leak → LeakCanary reference chain (what holds it); GC jank → allocation profiler on the hot path (Ch 90); early kills → trim handling (Ch 44); low-end crashes → footprint by device class (Ch 85).
- **Alerting:** OOM-rate spike (release regression, Ch 83); footprint regression release-over- release (Ch 85); low-memory-kill-rate spike (Ch 44).
- **Dashboards:** OOM rate + footprint percentiles by device/release; trim-kill rate; GC/jank correlation (Ch 90); top memory consumers.

89.16 System Design Interview Preparation

> ≥40 questions (Complex).

Fundamentals

1. *Managed heap?* ART heap + GC + per-app limit.
2. *What's a leak?* Reachable-from-GC-root object that should be freed.
3. *OOM cause?* Exceeding the per-app limit (bitmaps, unbounded caches).
4. *GC jank?* Per-frame allocations → GC pauses → dropped frames (Ch 90).
5. *Why memory matters?* Jank, OOM crash, background kill.

Leak taxonomy

6. *#1 Android leak?* Activity/View/Context held by static/singleton (Ch 46).
7. *Context fix?* applicationContext, not Activity.
8. *Listener leak?* Not unregistering; pair register/unregister (Ch 44).
9. *Coroutine leak?* GlobalScope/wrong scope; use viewModelScope (Ch 3/44).
10. *Inner class leak?* Implicit outer ref; static + WeakReference.
11. *Handler leak?* Delayed Runnable pins Activity; removeCallbacks.

Fixes

12. *Master fix?* Lifecycle scoping (Ch 44/3).
13. *WeakReference?* Unavoidable back-ref — doesn't pin host.
14. *repeatOnLifecycle?* Collect only while started — no leak (Ch 44).
15. *Unregister where?* Paired lifecycle callback (onStop/onDestroy).

Bitmaps/caches

16. *Why bitmaps?* Largest objects (w×h×4).

17. *Avoid bitmap OOM?* Downsample to view size; managed loader (Ch 59).
18. *Bounded cache?* LruCache size/count eviction.
19. *Size cache by?* Fraction of maxMemory.
20. *Image loader?* Coil/Glide — bounded + lifecycle (Ch 59).

Memory pressure

21. *onTrimMemory?* OS pressure signal; release non-critical (Ch 44).
22. *Trim levels?* UI_HIDDEN, RUNNING_LOW, COMPLETE.
23. *Why respond?* Avoid OOM + killed less eagerly (Ch 44).
24. *onLowMemory?* Older callback; evict caches.

Allocation/GC

25. *Per-frame alloc?* Hoist out of onDraw/recompose (Ch 90).
26. *Autoboxing?* Avoid on hot path.
27. *GC types?* ART generational; pauses drop frames.

Detection

28. *LeakCanary?* Debug — detects retained objects + reference chain.
29. *Memory Profiler?* Footprint, allocations, heap dumps.
30. *Field OOM?* Vitals/Crashlytics (Ch 85/83).
31. *Heap dump?* Snapshot of reachable objects; compare.

Edge cases

32. *Rotation leaks?* New Activity each time → multiply; LeakCanary (Ch 44).
33. *ViewModel leak?* Shouldn't hold View/Activity; survives rotation (Ch 44).
34. *Static Context?* Never; app-level if needed (Ch 46).
35. *Fragment view leak?* Null out view binding in onDestroyView.

Production

36. *OOM spike?* Release regression — bound/downsample (Ch 83).
37. *Footprint regression?* Profiler + field percentiles (Ch 85).
38. *Early bg kill?* Honor trim (Ch 44).
39. *Low-end crashes?* Footprint by device class (Ch 85).
40. *Sensitive data?* Minimize/clear; PII-safe dumps (Ch 28).
41. *Test leaks?* LeakCanary instrumented + heap assertions.
42. *StrictMode?* Flags leaked closeables/activities.

Design-change challenges

1. **Fix an Activity leak from a singleton.** applicationContext / WeakReference / unregister (Ch 46/44).
2. **Stop OOM in an image gallery.** Downsample + bounded Coil cache + trim (Ch 59/44).
3. **Eliminate GC jank in a custom view.** Hoist per-frame allocations (Ch 90).
4. **Keep the app resident longer in background.** Honor onTrimMemory (Ch 44).
5. **Catch leaks before they ship.** LeakCanary in CI + field OOM monitoring (Ch 85/83).

Mock walkthrough (abridged)

Clarify (leak or bloat? device targets? detection?) → heap/GC/limit model → leak taxonomy (static/singleton/coroutine/listener/inner, Ch 46/3/44) → fix by lifecycle scoping + applicationContext/WeakReference + unregister (Ch 44/46) → bound caches + downsample bitmaps (LruCache, Coil, Ch 59) → honor onTrimMemory (Ch 44) → avoid per-frame allocations (Ch 90) → detect: LeakCanary (dev) + Memory Profiler + field OOM (Ch 85/83) → observability (OOM rate + footprint by device).

89.17 Practical Exercises

1. **Beginner.** Add LeakCanary, reproduce an Activity leak (singleton holding `Activity` Context), and fix it with `applicationContext` (Ch 46). *Eval:* leak reported then gone.
2. **Intermediate.** Replace an unbounded bitmap cache with a bounded `LruCache` + Coil downsampling (Ch 59) and add `onTrimMemory` release (Ch 44). *Eval:* bounded; releases on trim.
3. **Advanced.** Scope a leaking `GlobalScope` coroutine to `viewModelScope` / `repeatOnLifecycle` (Ch 3/44) and null out a Fragment view binding in `onDestroyView`. *Eval:* no retained coroutine/view.
4. **Production debugging.** Investigate an OOM spike with a heap dump (find the top retained objects); investigate GC jank with the allocation profiler (Ch 90). *Eval:* root cause found.
5. **System design challenge.** Design memory budgets by device class + CI leak detection (instrumented LeakCanary) + field OOM monitoring (Ch 85/83). *Eval:* gated, device-tiered, observed.

(Solutions withheld unless requested.)

89.18 Chapter Summary & Reference Sheet

Key decisions: keep memory healthy under ART's **managed heap + GC + per-app limit** by treating a **leak** as any object still **reachable from a GC root** (a static/singleton field, a live thread/coroutine, a registered listener) that **should** be freed — most often an **Activity/View/Context** captured by a **singleton or static** (Ch 46), a **coroutine in the wrong scope** (`GlobalScope`, Ch 3), a **non-static inner class**, a `Handler` / `Runnable`, or an **unregistered listener/receiver** that outlives its **lifecycle** (Ch 44); **prevent leaks** by **scoping every reference to a lifecycle** (`viewModelScope` / `lifecycleScope` + `repeatOnLifecycle` / `collectAsStateWithLifecycle`, Ch 3/44), **unregistering** listeners and **canceling** coroutines, **never holding Activity/View/Context in long-lived objects** (use `applicationContext`, or a `WeakReference` for unavoidable back-refs, Ch 46), **bounding every cache** (`LruCache` by size/count) and **image memory** with **downsampling** (Ch 59), and **responding to** `onTrimMemory` by releasing non-critical memory so the app **avoids OOM** (Ch 83) and is **killed less eagerly** in the background (Ch 44); **avoid per-frame/per- recomposition allocations** that churn the GC into frame-dropping **pauses** (Ch 90); and **detect** problems with **LeakCanary** (dev — gives the retaining reference chain), the **Memory Profiler** (footprint + heap dumps), and **field OOM/trim monitoring** (Ch 85/83) — because leaks and bloat surface as **jank, OOM crashes, and background kills**.

Concepts: managed heap + GC + per-app limit, reachability/GC roots, leak taxonomy (static/singleton/coroutine/listener/inner/Handler), lifecycle scoping (Ch 44/3), `applicationContext` / `WeakReference` (Ch 46), bounded caches (`LruCache`) + bitmap downsampling (Ch 59),

`onTrimMemory` / `ComponentCallbacks2` (Ch 44), allocation discipline → GC jank (Ch 90), LeakCanary + Memory Profiler + field OOM (Ch 85/83).

Common mistakes: holding `Activity` / `View` / `Context` in a singleton/static; `GlobalScope` coroutines capturing lifecycle objects; not unregistering listeners/receivers; unbounded caches; hand-decoded bitmaps without downsampling (OOM); ignoring `onTrimMemory`; per-frame allocations (GC jank); not nulling Fragment view bindings; finding leaks only in production (no LeakCanary).

Checklist:

- [ ] lifecycle-scoped coroutines/collection (viewModelScope, repeatOnLifecycle, Ch 3/44)
- [ ] unregister listeners/receivers in paired callback (Ch 44)
- [ ] applicationContext (not Activity) in long-lived objects; WeakReference for back-refs (Ch 46)
- [ ] bounded caches (LruCache) + downsampled bitmaps / Coil (Ch 59)
- [ ] honor onTrimMemory → release non-critical (Ch 44)
- [ ] no per-frame/recomposition allocations (Ch 90)
- [ ] null Fragment view bindings in onDestroyView
- [ ] LeakCanary in debug
- [ ] Memory Profiler footprint audits
- [ ] field OOM + trim monitoring (Ch 85/83).

Key APIs: `viewModelScope` / `lifecycleScope` / `repeatOnLifecycle` / `collectAsStateWithLifecycle` (Ch 3/44), `@ApplicationContext` / `applicationContext` (Ch 46), `java.lang.ref.WeakReference`, `android.util.LruCache` (`sizeOf` / `trimToSize` / `evictAll`), `ComponentCallbacks2.onTrimMemory` (Ch 44), `Handler.removeCallbacks`, Coil/Glide managed caches (Ch 59), LeakCanary (debug), Android Studio Memory Profiler + heap dumps, StrictMode, field OOM via vitals/Crashlytics (Ch 85/83).

References:

- Manage your app's memory: https://developer.android.com/topic/performance/memory
- onTrimMemory / ComponentCallbacks2: https://developer.android.com/reference/android/content/ComponentCallbacks2
- LeakCanary: https://square.github.io/leakcanary/
- Memory Profiler: https://developer.android.com/studio/profile/memory-profiler
- (composes) Ch 44 Lifecycle, Ch 3 Threading, Ch 59 Image Loading, Ch 46 DI, Ch 85 Perf, Ch 83 Crash/OOM, Ch 90 Jank

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/pressure/failure diagrams; lifecycle-scope + safe-context + WeakReference + bounded-cache + trim + allocation code; concurrency (lifecycle master fix, rotation compounding, bounded+trim); failure (scope+bound+ trim+detect principle); 5 levels; scoping-vs-manual, appContext-vs-Activity, managed-loader trade-offs; security (clear sensitive, PII-safe dumps); perf (allocations, bound, downsample, trim); cache/trim/leak tests; observability (OOM + footprint by device); 42 Q&A + 5 challenges. ✅
- **Expand later:** lifecycle (Ch 44), threading (Ch 3), image loading (Ch 59), jank (Ch 90), crash/OOM (Ch 83).
- **Interview coverage:** 42 Q&A — exceeds the ≥40 Complex bar.